Group 09

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1. INTRODUCTION

This project is motivated by the TREC-COVID Shared Task where users of the system are expected to be "Researchers, clinicians, and policy makers involved with the response to COVID-19 are constantly searching for reliable information on the virus and its impact." The first phase of the project encompasses exploration of text processing options, implementation of an IR system with boolean, vector space and probabilistic retrieval models, exploration of ranking, and evaluation of the implemented IR system. During the first phase of the project, documents are ranked according to their metadata, which includes both their title and abstract sections.

2. DATASET EXPLORATION

The original dataset is composed of 192509 documents with 1195 duplicate ids, 812572 distinct terms and an average document length of 153.63 terms (when considering title + abstract). After handling the repeated documents (by concatenating all of their characteristics), our dataset contains 191120 distinct documents. On average, each document of the resulting dataset is composed of 107.31 terms, with the largest document having 16472 terms and the smallest one 1 term. By checking the documents content we were able to access that there are 197892 unique terms being the most frequent one being "patient", appearing a total of 206088 times, and the least frequent one "lowprevalence" appearing only 1 time in total. The 10 top terms in the documents, before and after preprocessing, are presented in **Figure 1** and **Figure 2**.

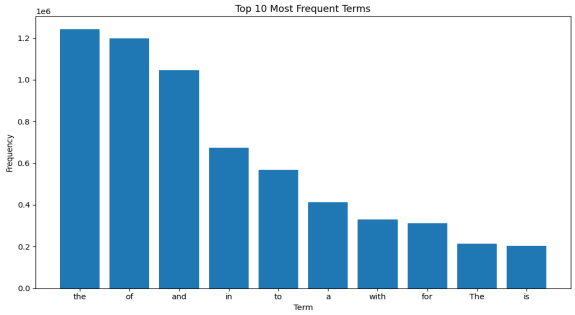


Figure 1. Top 10 terms on corpus before preprocessing

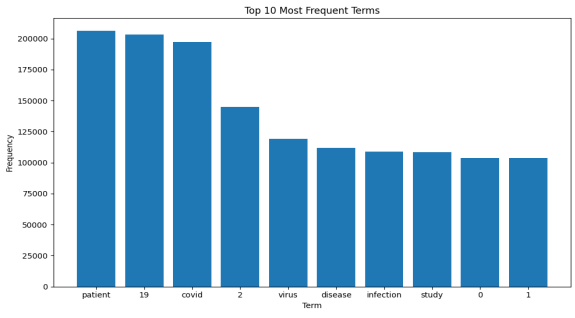


Figure 2. Top 10 terms corpus after preprocessing

Regarding the queries, the original dataset contains 50 queries and 102 unique terms, averaging a length of 3.24 terms, where the longest query contains 7 terms and the smaller one 2 terms. After preprocessing the queries, the quantity of unique terms gets reduced to 88 terms, and, consequently, the biggest query presents now a total of 5 terms, the smaller one 2 queries and, on average the queries now present a length of 2.98 terms. The difference between the top 10 terms before and after preprocessing can be seen on **Figure 3** and **Figure 4**.

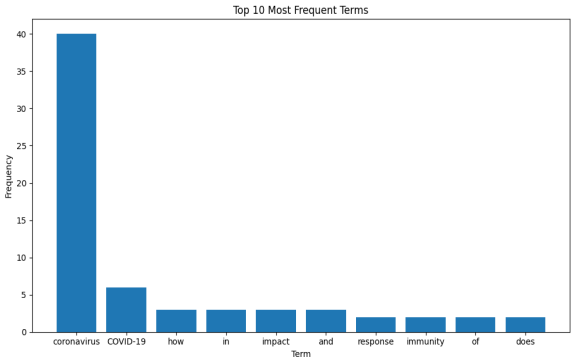


Figure 3. Top 10 terms on queries (Q) before preprocessing

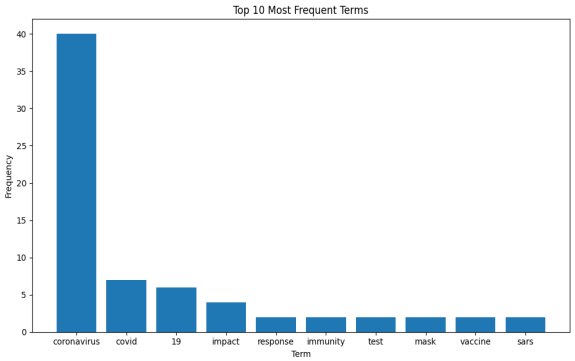


Figure 4. Top 10 terms on queries (Q) after preprocessing

3. IR SYSTEM PERFORMANCE

All the execution time and memory space results were obtained from the same machine.

3.1 Processing Performance

Five text processing options were implemented:

- Stopword Removal;
- Normalization;
- Tokenization;
- Lemmatization;
- Stemming (Porter);

All the possible combinations of these options were tested, and the resulting metrics (MAP, nDCG, nDCG@10 and P@10) were analyzed and compared. This led to the conclusion that performing removal of stopwords followed by normalization (lowercasing and non-alphanumeric character removal) and, despite its higher computational cost, lemmatization delivered the highest retrieval effectiveness.

This preprocessing step resulted in **90.7276 seconds** of processing time, and the processed documents occupied **180.86 MB**.

Table 1. Time taken for Processing stage

Stage	Time taken (s)	Space used (MB)
Processing	90.7276	180.86

3.2 Indexing Performance

The inverted index is built using a term-document frequency dictionary, where each term is mapped to a dictionary of documents and their respective term frequencies (TF). The indexing process involved iterating through the dataset, preprocessing the text (as described in the processing phase), and populating the index structure.

This resulted in **9.1878 seconds** to process the dataset, and the final index occupied **1378.98 MB**.

Table 2. Time and Space for Indexing Stage

Stage	Time taken (s)	Space used (MB)
Indexing	9.1878	1378.98

3.3 Retrieval Performance

3.3.1 Model Performance

- *Boolean Model*

In our evaluation of the Boolean retrieval model, we chose to use only the query title for matching. Including the query description and narrative would significantly reduce the number of matches since the Boolean model relies on strict set-based operations. Given its binary nature (where documents must contain all query terms to be retrieved), adding more text would make matches rarer, leading to poorer retrieval performance. The metrics obtained for this model, and the corresponding distributions, are presented in **Table 3** and **Figure 5**, respectively.

Table 3. Mean metrics for Boolean over Q with p=1000

Precision	Recall	F-measure with β=1	Time Per Query (s)
0.3502	0.2205	0.2144	0.0061

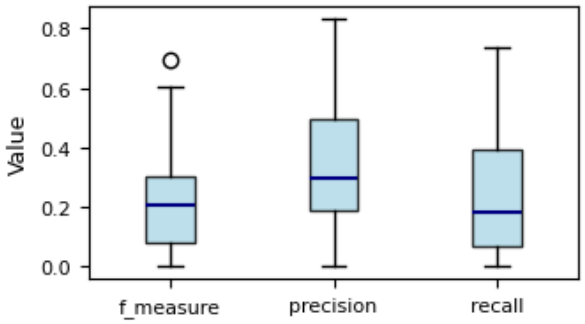


Figure 5. Metric Distributions for Boolean over Q

- *TF-IDF*

The TF-IDF model exhibits many variants, and different weightings are allowed for queries and documents. A few weighting schemes were tested, and the one that yielded the best results metric-wise was **lnn.ntn**. The metrics obtained for this model, and the corresponding distributions, are presented in **Table 4** and **Figure 6**, respectively.

Table 4. Mean metrics for TF-IDF over Q with p=1000

MAP	nDCG	nDCG@10	P@10	Time Per Query (s)
0.1878	0.3878	0.5429	0.6380	0.1208

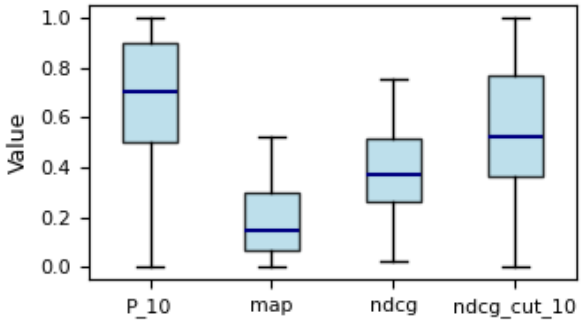


Figure 6. Metric Distributions for TF-IDF over Q with p=1000

- *BM25*

The BM25 model exhibits two parameters, k1 and b, which control the term frequency scaling and the document length normalization, respectively. By trying all the combinations for the values, with k1 between 1 and 2 with a step of 0.1 and b between 0 and 1 with a step of 0.1, it was possible to conclude that **k1=1.1** and **b=0.4** yielded the best results metric-wise. The metrics obtained for this model, and the corresponding distributions, are presented in **Table 5** and **Figure 7**, respectively.

Table 5. Mean metrics for BM25 over Q with p=1000

MAP	nDCG	nDCG@10	P@10	Time Per Query (s)
0.2148	0.4153	0.6456	0.7320	0.1259

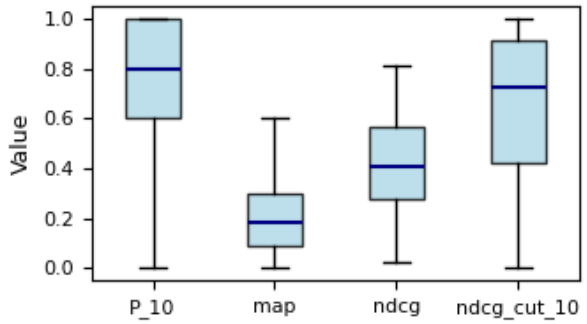


Figure 7. Metric Distributions for BM25 over Q with p=1000

- *Reciprocal Rank Fusion (TF-IDF & BM25)*

Reciprocal Rank Fusion (RRF) can be used for placing consensus between the ranking outputs of TF-IDF and BM25. By leveraging the strengths of each model, RRF helps produce a more balanced ranking that is closer to an optimal retrieval performance. Given a set of queries, we retrieved ranked document lists from each scoring function and fused them using RRF. The metrics obtained for this model, and the corresponding distributions, are presented in **Table 6** and **Figure 8**, respectively.

Table 6. Mean metrics for RRF over Q with *p*=1000

MAP	nDCG	nDCG@10	P@10	Time Per Query (s)
0.2151	0.4299	0.6269	0.7140	0.2453

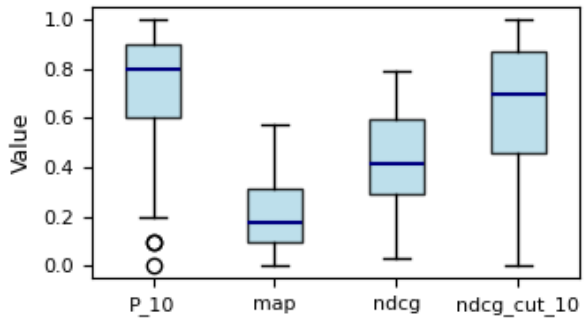


Figure 8. Metric Distributions for RRF over Q with p=1000

3.3.2 *Model Comparison*

While the precision and recall obtained for the unranked retrieval model, Boolean, can be perceived as unsatisfactory, when comparing it with one of the ranked retrieval models, like BM25, it is possible to conclude that its performance is actually decent, as shown in **Table 7**, while being much faster at retrieval. Its biggest drawback is the lack of ranking, which makes any of the other models more desirable. Experimentations were conducted to evaluate the performance of first performing a Boolean retrieval followed by a ranked retrieval on the previously retrieved documents, but the low recall exhibited by the Boolean model hindered the overall retrieval performance.

Table 7. Comparison between Boolean and BM25 over Q

	p	Precision	Recall
Boolean	[redacted]	0.3502	0.2205
BM25	1000	0.2094	0.3970

Regarding the ranked retrieval models, it is evident that both BM25 and RRF lead to a better performing IR system. Between BM25 and RRF, we can observe that the RRF model slightly improved MAP and nDCG over BM25, indicating better overall ranking quality. However, nDCG@10 and P@10 are slightly lower, suggesting a reduced effectiveness in top-ranked retrieval. The boxplots further reveal that while RRF achieves a higher median, its lower quartile is significantly worse than BM25, suggesting that some queries perform poorly under this approach. In terms of efficiency, though, this was clearly the lengthiest approach, which is not surprising given that it works as a combination of previous models.

3.3.3 *Best Performing Model - BM25*

Depending on the user's preference or use case, more importance is given to either precision or recall. The developed IR system was tested for a range of *p* (number of retrieved documents), and both precision and recall metrics were computed, which are presented in **Table 8**. As expected, if the objective of the user is precision, and thus minimizing false positives, then a lower *p*, under 100, is optimal. If the user's intention is recall, and thus minimizing false negatives and maximizing true positives, then a higher *p* is recommended, preferably equal or higher than 5000. However, it is also possible to observe that, as *p* increases, increase in recall progressively slows down.

Table 8. Precision and Recall with varying p

p	Precision	Recall	FP	FN	TP
10	0.7320	0.0166	2.7	526.0	7.3
100	0.5248	0.1121	47.5	480.8	52.5
500	0.3084	0.3029	345.8	379.1	154.2
1000	0.2094	0.3970	790.6	323.8	209.4
5000	0.0639	0.5791	4680.7	214.0	319.3

From the obtained metrics for this model, **Table 5**, and more notably the relatively low MAP and the relatively high nDCG@10 values obtained, it is possible to conclude that the developed IR System is more adequate for precision purposes. As for the distribution of each metric, **Figure 7**, they exhibit high variance, meaning that the performance of retrieval/ranking does vary considerably between queries.

4. Conclusions

While all the objectives were met, due to time constraints, the team was unable to fully explore the design of the IR system. Given the domain specific nature of the corpus, the target users and the obtained results, we believe that the current IR System is not the most adequate and could be greatly improved. For the end users, a system with higher MAP and nDCG metrics is likely to be preferable, since they would benefit from the novelty factor of the retrieved documents. Furthermore, the variability of performance between queries should be mitigated. We hope to address these flaws going into the next phase of the project.